MIL-E-8970(ASG)
9 OCTOBER 1967

MILITARY SPECIFICATION

ENGINES AND RELATED PROPULSION AND POWER EQUIPMENT,
AIRCRAFT, ACCEPTANCE TESTS OF, SAMPLING PLAN FOR, STATISTICAL

This specification has been approved by the Department of the Air Force and by the Naval Air Systems Command.

1. SCOPE

1.1 This specification covers acceptance tests, by the statistical sampling plan (SSP), for all types of production aircraft engines or any related propulsion and power equipment requiring tear-down after the initial run. All references herein to equipment shall be construed to mean the item to which the SSP is applied. Application of the SSP shall be to a specific equipment type and model. When agreed upon by the Government representative and the contractor, substantially identical equipments of different model designations may be grouped together and presented to the procuring Service as one series for SSP evaluation.

2. APPLICABLE DOCUMENTS

2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Office of Assistant Secretary of Defense

Handbook H-106 — Multi-level Continuous Sampling Procedures and Tables for Inspection by Attributes

(Copies of the above publication may be obtained from the Government Printing Office, Washington, D. C., 20402.)

3. REQUIREMENTS

3.1 Preliminary requirement.- A detailed study of the equipment shall be evaluated for statistical sampling and reported to the procuring Service. The report shall contain a detailed analysis of test data, tear-down and rebuild inspection records, photographs and other pertinent data. Defects will be assigned to one of the classes defined in section 4 as determined by the procuring Service. A review and approval of the report by the procuring Service is required before consideration can be given to the use of SSP for acceptance of the equipment.

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3.4.1 Records.- The contractor shall maintain a detailed record of inspection results for all equipments by model and serial number which have been processed under this sampling procedure. Such records shall be readily available to the Government representative.

3.2 Initiation of SSP.- At least 125 consecutive equipments shall be torn down and inspected. To qualify for sampling, not more than two defectives (see 4.1.4) are permitted in a series of 75 consecutive equipments and no defectives in a subsequent series of 50. Finding a defective during the last 50 equipments requires the initiation of a new series of 50. A major change which, in the opinion of the procuring Service, affects the substantially identical nature of the equipment, requires suspension and reinitiation of the SSP.

3.3 Minimum production.- A minimum production of 16 equipments per month is required before the sampling program may be initiated. At least this production shall be maintained to continue the SSP. If the average production for a 6-month period falls below 16 equipments per month, or if less than 10 equipments are built in any month, SSP shall be suspended. At such time as the minimum production requirements are met, sampling may be resumed at level I when 50 consecutively produced units have been inspected and no defectives have been found.

3.4 Responsibility for inspection.- Statistical samples shall be selected by the Government representative on a random basis in accordance with "random sampling procedures" described in Handbook H-106. A mechanical method of random sample selection shall be used. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may utilize his own facilities or any other facility acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

3.5 Sampling levels.- The following sampling levels are established by this specification (see table I):

TABLE I. Sampling levels

Level	Sampling Frequency	Equipments torn-down	Acceptance number
I	1/2	75	1
II	1/4	75	1
III	1/10	75	1

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3.5.1 Level I.- The contractor shall tear-down and inspect 50 percent of the equipments until 75 have been torn down and inspected. If not more than one defective is found proceed to level II. Corrective action shall be taken when a defective is found, but the sampling may continue. Upon the detection of a second defective in any consecutive group of 75 inspected equipments, corrective action shall be taken, and the sampling procedure shall be suspended, until 50 consecutive equipments are found free of defectives. Sampling shall then be resumed at level I. If a defective is found during the inspection of the 50 consecutive equipments the statistical sampling plan shall remain suspended pending corrective action satisfactory to the procuring Service. No less than 50 consecutive equipments shall be found free of defectives before reinitiation of sampling at level I.

3.5.2 Level II.- The contractor shall tear-down and inspect 25 percent of the equipments until 75 have been torn down and inspected. If not more than one defective is found proceed to level III. Corrective action shall be taken when a defective is found, but the sampling may continue. Upon the detection of a second defective in any consecutive group of 75 inspected equipments corrective action shall be taken, and the SSP shall be suspended until 50 consecutive equipments are found free of defectives. Sampling shall then be resumed at level I. If a defective is found during the inspection of the 50 consecutive equipments, the SSP shall remain suspended pending corrective action satisfactory to the procuring Service. No less than 50 consecutive equipments shall be found free of defectives before reinitiation of sampling at level I.

3.5.3 Level III.- The contractor shall tear-down and inspect 10 percent of the equipments. If not more than one defective is found in any 75 consecutively inspected equipments, corrective action shall be taken and inspection may be continued at level III. Upon the detection of a second defective in any consecutive group of 75 inspected equipments, corrective action shall be taken and the SSP shall be suspended until 50 consecutive equipments are found free of defectives. Sampling shall then be resumed at level I. If a defective is found during the inspection of the 50 consecutive equipments the SSP shall remain suspended pending corrective action satisfactory to the procuring Service. No less than 50 consecutive equipments shall be found free of defectives before reinitiation of sampling at level I.

3.6 Progression and retrogression.- The flow chart of figure 1 shows the changes in levels of sampling and the criteria for progression or retrogression. It is intended to supplement the procedure described above. Retrogression to the next lower level of sampling is also required if the number of class III defects totals 10 or more in any consecutive group of 75 inspected equipments. If there is a class I or II defect during a sampling level, the effect of a number of class III defects in addition to the class I or II defect will be taken into consideration by the procuring Service in determining need for retrogression to a lower sampling level.

3.6.1 Failures in service.- If at any time it is determined that there has been a significant lowering of quality based on reports of equipment failures in service, the quality assurance representative (QAR) will direct suspension of sampling and notify the procuring Service of his action. In the absence of action by the QAR, the procuring Service shall have the prerogative of initiating action to suspend sampling.

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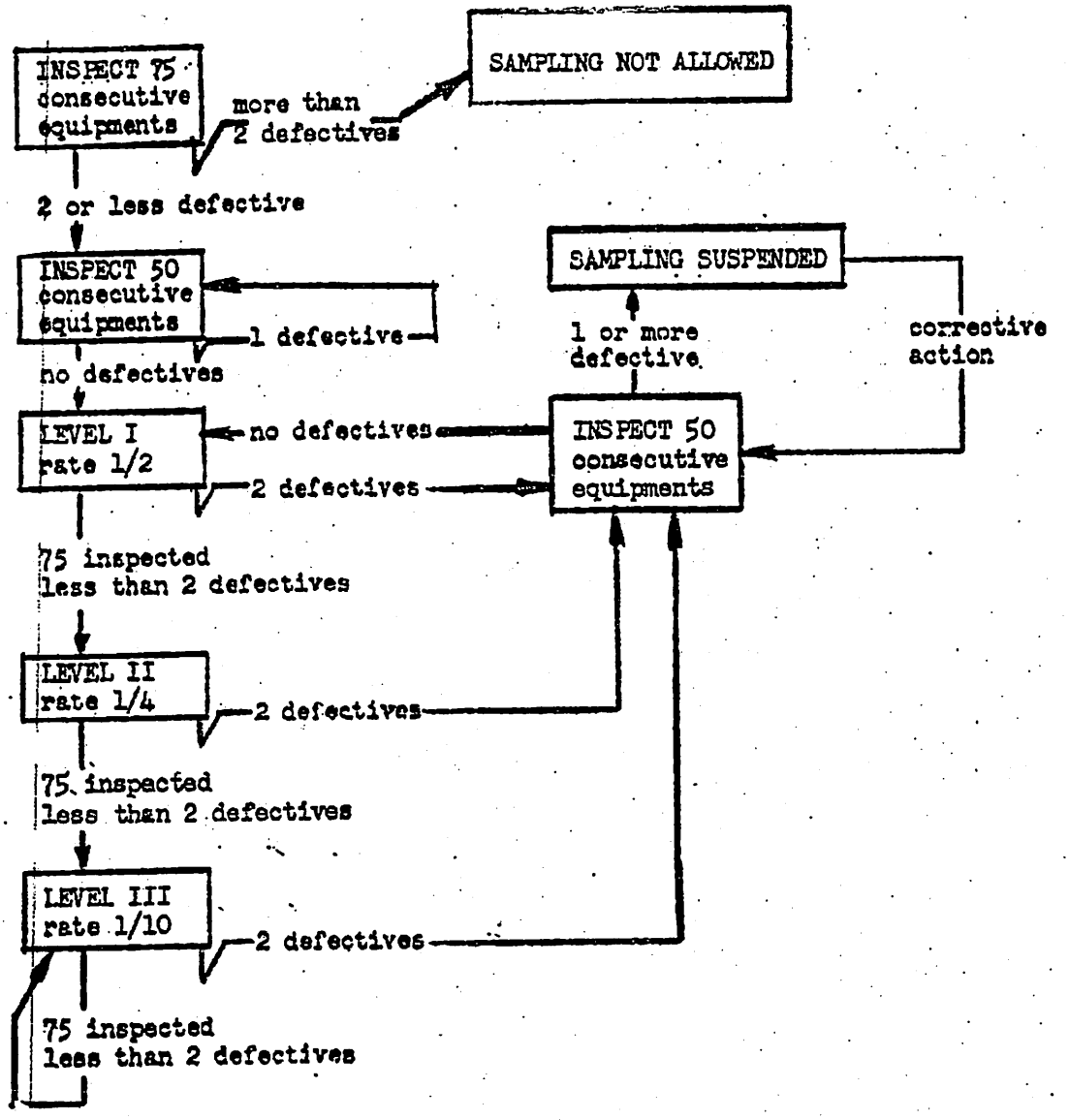


FIGURE 1. SSP flow process chart

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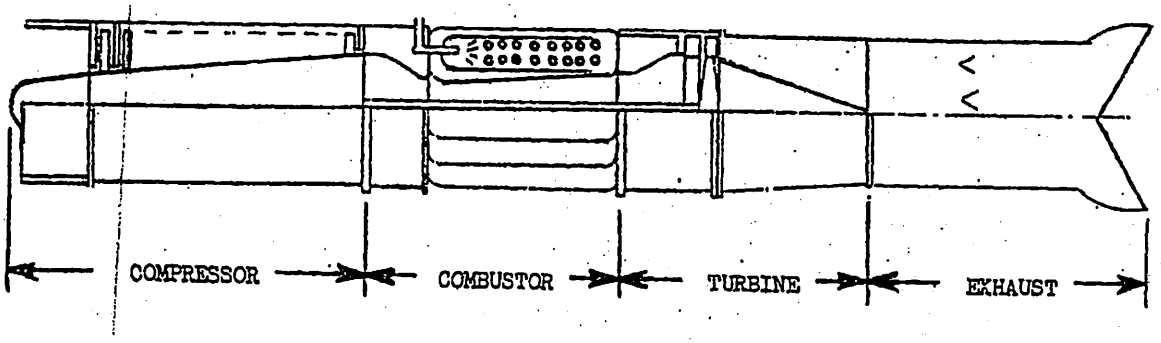


FIGURE 2. Typical equipment major subassemblies

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3.7 Optional sampling procedures.-

3.7.1 H-106 sampling.- After qualification for SSP, the contractor may employ in lieu of the sampling plan described herein as approved by the Government, the use of the appropriate equivalent continuous sampling procedure as contained in the DOD Inspection and Quality Control Handbook (Interim) H-106, "Multi-level Continuous Sampling Procedures and tables for Inspection by Attributes." When employed for equipment sampling, the optional plan shall conform to an AOQL of 1.0 percent.

3.7.2 Subassembly sampling.- Following qualification for SSP (see 3.2), upon request by the contractor and approval by the procuring Service, the SSP may be applied to major subassemblies (see figure 2). Exact designation of major subassemblies will be made by the contractor, subject to the approval of the procuring Service. All reference to equipment shall be construed to mean the subassembly to which this procedure is applied.

3.8 Adequate test facilities.- The contractor shall retain adequate test facilities to maintain his schedule rate of production with 100 percent inspection of equipments.

4. NOTES

4.1 Definitions.- For the purpose of this specification, the following definitions apply.

4.1.1 Class I defect.- Defect, which may cause equipment failure prior to the specified repair/overhaul schedule, which can be found only by disassembly of the equipment.

4.1.2 Class II defect.- Defect which may cause equipment failure prior to the specified repair/overhaul schedule set forth in the appropriate maintenance instructions, but which may be detectable without disassembly of the equipment. Such a defect should be one which can be detected or seen from the outside of the equipment.

4.1.3 Class III defect.- Defect which would not cause equipment failure prior to the specified repair/overhaul schedule.

4.1.4 Defective.- An equipment which is found to have one or more class I or two or more class II defects.

4.1.5 Procuring Service.- The Service which has the responsibility for procurement of the equipment.

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Navy - AS
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Preparing activity:
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